CSA06507-Generative AI

Lab Experiment's

.

Unit 2: Lab Programs for Prompting

Experiment 4: Professional Email Generation Using Zero-Shot, One-Shot, and Few-Shot Prompting

1. Aim

To design Zero-Shot, One-Shot, and Few-Shot prompts to generate a professional email requesting leave due to illness and compare the generated emails based on tone, grammar, formatting, and completeness.

2. Algorithm

1. Start the program.
2. Define the task of writing a professional sick-leave email.
3. Create a Zero-Shot prompt without examples.
4. Create a One-Shot prompt with one example email.
5. Create a Few-Shot prompt with two example emails.
6. Generate an email for each prompting technique.
7. Compare the emails based on tone, grammar, formatting, and completeness.
8. Display all emails and comparison results.
9. Stop the program.

3. Program (Google Colab)

Dear Sir/Madam,

I am writing to inform you that I am unwell

and unable to attend today. I kindly request

leave for one day to recover.

Thank you for your understanding.

Yours sincerely,

[Your Name]

"""

email_one = """

Subject: Sick Leave Request

Dear Sir/Madam,

to attend college today. I kindly request you

```
to grant me leave for one day.
Thank you for your consideration.
Yours sincerely,
[Your Name]
"""
email_few = """
Subject: Request for Sick Leave
Dear Sir/Madam,
I am feeling unwell and have been advised to
rest. Therefore, I request leave for one day
to recover. I will complete any pending work
after returning.
Thank you for your understanding.
Yours sincerely,
[Your Name]
"""
# Display prompts and emails
print("ZERO-SHOT PROMPT:")
print(zero_shot)
print("GENERATED EMAIL:")
print(email_zero)
print("\nONE-SHOT PROMPT:")
print(one_shot)
print("GENERATED EMAIL:")
print(email_one)
print("\nFEW-SHOT PROMPT:")
print(few_shot)
print("GENERATED EMAIL:")
print(email_few)
# Compare the emails
print("\nCOMPARISON")
```

```
print("Criteria       Zero-Shot   One-Shot   Few-Shot")
print("Tone           Good        Good       Very Good")
print("Grammar        Good        Good       Good")
print("Formatting     Good        Good       Very Good")
print("Completeness   Good        Good       Very Good")
```

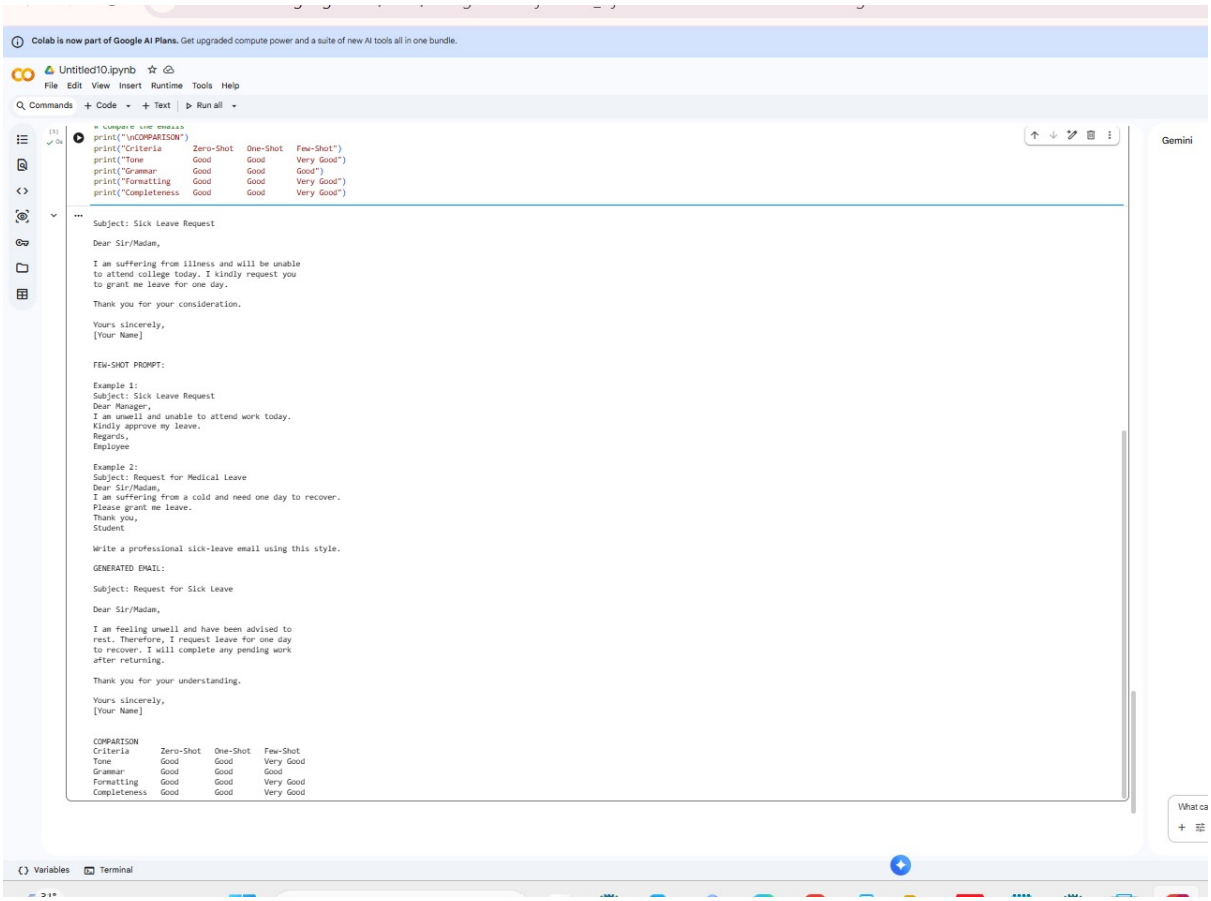
4. Input

Task: Generate a professional email requesting leave due to illness.

Requirements:

- Include a suitable subject line.
- Use a polite greeting.
- Explain the reason for leave.
- Request leave for one day.
- Include a professional closing.

5. Output



6. Comparison Table

Criteria	**Zero-Shot**	**One-Shot**	**Few-Shot**
Tone	Polite	Professional	Professional and considerate
Grammar	Correct	Correct	Correct
Formatting	Well structured	Well structured	Well structured
Completeness	Includes main details	Includes main details	Includes leave request and follow-up plan

7. Result

Zero-Shot, One-Shot, and Few-Shot prompting techniques were demonstrated for generating professional sick-leave emails. The emails were compared based on tone, grammar, formatting, and completeness. Zero-Shot generates an email directly, One-Shot follows one example, and Few-Shot uses multiple examples to guide the writing style. The Few-Shot approach provides a more detailed email in this demonstration.

Note: This beginner-friendly Colab program displays predefined example emails to demonstrate the prompting techniques. It does not call an actual AI model API.